

Research Interests

Multi-party/Multi-modal Dialogue; Discourse/Semantic Parsing; AI for Education/Healthcare/Hydrology/Material Science etc.

Education

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|-------------|---|
| 2015 – 2022 | Doctor of Philosophy
Computer Science
<i>University of Utah</i> |
| 2009 – 2012 | Master of Science
Computer Science
<i>Huazhong University of Science and Technology (HUST)</i> |
| 2005 – 2009 | Bachelor of Engineering
Computer Science
<i>Huazhong University of Science and Technology (HUST)</i> |

Research Experience

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| University of Oklahoma, Norman | 2024 - Now |
| – I am an Assistant Professor in the School of Computer Science, affiliated with Data Science and Analytics Institute, leading the Natural Language Processing group [2, 3, 1, 4] | |
| University of Colorado Boulder, Boulder | 2022 - 2024 |
| – I was a post-doctoral researcher at University of Colorado Boulder, with Prof. James Martin and Prof. Martha Palmer, Boulder NLP Groups, and Institute of Cognitive Science. My research mainly covers conversational modelling [9, 6, 0], speech-aware models [8, 7], education data science on mathematic teaching and tutoring [2], small group collaborative learning [7]. | |
| Utah NLP Group, University of Utah, Salt Lake City | 2015 - 2022 |
| – I studied methods for predicting structured representations of natural language text, e.g., semantic parsing [12, 18], dialogue system [14, 10], self-supervised learning on database workloads [11], visualization for network pruning [5]. | |
| Intern at Amazon Web Services, Remote | Summer 2020 |
| – Our paper on schema-guided dialog got accepted to NAACL 2021 [10]. | |
| Intern at Amazon Web Services, Seattle | Summer 2019 |
| – Cross framework meaning representation parsing, ranked 1st on AMR on CoNLL 2019 shared task [12]. | |
| Intern at Tencent, Wechat, Palo Alto | Summer 2018 |
| – End-to-End Goal-oriented Dialogue System, ranked 2nd in DSTC7 track1 [14, 17] | |
| – Rhetorically Controlled Poetry Generation[13] got accepted to ACL 2019. | |
| CGCL Lab, HUST, Wuhan | 2008 - 2012 |
| – When I was a master student at HUST, I worked with Prof. Xia Xie and Prof. Hai Jin. My research are around statistical computing, programming language [15, 16], system virtualization [20, 19] | |

Teaching & Mentoring

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|-------------|------------|---|
| Fall 2025 | Instructor | Natural Language Processing (CS 5293) , U of Oklahoma |
| Spring 2025 | Instructor | Text Analytics (CS 5293) , U of Oklahoma |
| Fall 2023 | Instructor | Natural Language Processing (CSCI-LING 5832) , U of Colorado Boulder |

Publications

- [1] Yu Cai, Ziang He, Yanyan Luo, **Jie Cao**, Yuchen Liu, Zhengping Luo, and Shangqing Zhao. “Adversarial Attacks on Cooperative Spectrum Sensing: A LLM-Powered Multi-Agent Approach”. *2025 IEEE 26th International Workshop on Signal Processing and Artificial Intelligence for Wireless Communications (SPAWC)* (2025).
- [2] **Jie Cao**, Abhijit Suresh, Jennifer Jacobs, Charis Clevenger, Amanda Howard, Chelsea Brown, Brent Milne, Tom Fischhaber, Tamara Sumner, and James H. Martin. “Enhancing Talk Moves Analysis in Mathematics Tutoring through Classroom Teaching Discourse”. *The 31st International Conference on Computational Linguistics (COLING 2025)*. 2025.
- [3] Jannatun Naim, **Jie Cao**, Fareen Tasneem, Jennifer Jacobs, Brent Milne, James Martin, and Tamara Sumner. “Towards Actionable Pedagogical Feedback: A Multi-Perspective Analysis of Mathematics Teaching and Tutoring Dialogue”. *Proceedings of the 18th International Conference on Educational Data Mining, 2025* (2025).
- [4] Yingheng Tang, Wenbin Xu, **Jie Cao**, Jianzhu Ma, Weilu Gao, Steve Farrell, Benjamin Erichson, Michael W Mahoney, Andy Nonaka, and Zhi Yao. “MatterChat: A Multi-Modal LLM for Material Science”. *arXiv preprint arXiv:2502.13107* (2025).
- [5] Zhimin Li, Shusen Liu, Xin Yu, Kailkhura Bhavya, **Jie Cao**, Diffenderfer James Daniel, Peer-Timo Bremer, and Valerio Pascucci. “Understanding Robustness Lottery”: A Geometric Visual Comparative Analysis of Neural Network Pruning Approaches”. *IEEE Transactions on Visualization and Computer Graphics* (2024).
- [6] Jon Z Cai, Brendan King, Margaret Perkoff, Shiran Dudy, **Jie Cao**, Marie Grace, Natalia Wojarnik, Ananya Ganesh, James H Martin, Martha Palmer, et al. “Dependency Dialogue Acts–Annotation Scheme and Case Study”. *The 13th International Workshop on Spoken Dialogue Systems Technology* (2023).
- [7] **Jie Cao**, Ananya Ganesh, Jon Cai, Rosy Southwell, E Margaret Perkoff, Michael Regan, Katharina Kann, James H Martin, Martha Palmer, and Sidney D’Mello. “A Comparative Analysis of Automatic Speech Recognition Errors in Small Group Classroom Discourse”. *Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization*. 2023, pp. 250–262.
- [8] Ananya Ganesh, **Jie Cao**, E. Magerate Perkoff, Rosy Southwell, Martha Palmer, and Katharina Kann. “Mind the Gap between the Application Track and the Real World”. *Proceedings of the 61th Annual Meeting of the Association for Computational Linguistics, 2023* (2023).
- [9] E. Margaret Perkoff, Abhidip Bhattacharyya, Jon Cai, and **Jie Cao**. “Comparing Neural Question Generation Architectures for Reading Comprehension”. *18th Workshop on Innovative Use of NLP for Building Educational Applications, 2023* (2023).
- [10] **Jie Cao** and Yi Zhang. “A Comparative Study on Schema-Guided Dialogue State Tracking”. *Proceedings of NAACL 2021*. 2021.
- [11] Debjyoti Paul*, **Jie Cao***, Feifei Li, and Vivek Srikumar. “Database Workload Characterization with Query Plan Encoders”. *Proc. VLDB Endow.* 15.4 (2021), pp. 923–935. issn: 2150-8097. doi: [10.14778/3503585.3503600](https://doi.org/10.14778/3503585.3503600). url: <https://doi.org/10.14778/3503585.3503600>.
- [12] **Jie Cao**, Yi Zhang, Adel Youssef, and Vivek Srikumar. “Amazon at MRP 2019: Parsing Meaning Representations with Lexical and Phrasal Anchoring”. *Proceedings of the Shared Task on MRP at the CoNLL 2019*. 2019.
- [13] Zhiqiang Liu, Zuohui Fu, **Jie Cao**, Gerard de Melo, Yik-Cheung Tam, Cheng Niu, and Jie Zhou. “Rhetorically Controlled Encoder-Decoder for Modern Chinese Poetry Generation”. *Proceedings of ACL 2019*. Florence, Italy, 2019.
- [14] Shuo Sun*, Yik-Cheung Tam*, **Jie Cao***, Can-Xiang Yan, Zuohui Fu, Cheng Niu, and Jie Zhou. “End-to-end Gated Self-attentive Memory Network for Dialog Response Selection”. *Proceedings of AAAI, DSTC7 Workshop*. 2019.
- [15] Xijiang Ke, Hai Jin, Xia Xie, and **Jie Cao**. “A distributed SVM method based on the iterative MapReduce”. *International Conference on Semantic Computing(ICSC)*. IEEE. 2015.
- [16] Xia Xie, **Jie Cao**, Hai Jin, Xijiang Ke, and Wenzhi Cao. “JRBridge: A framework of large-scale statistical computing for R”. *Asia-Pacific on Services Computing Conference (APSCC)*. IEEE. 2012.

Thesis & Dissertation & Patents

- [17] TAN Yizhang, Shuo Sun, **Jie Cao**, Le Tian, Cheng Niu, and Jie Zhou. “Encoding method, apparatus, and storage medium”. Pat. US Patent 11,934,788. 2024.
- [18] **Jie Cao**. “Inductive Biases for Deep Linguistic Structured Prediction with Independent Factorization”. University of Utah, 2022.
- [19] **Jie Cao**, Xia Xie, and Jin Hai. “A real-time scheduling system on embeded virtualization”. Patent ZL201110410689.1 (CN). 2011.
- [20] **Jie Cao**. “Cache Performance Evaluation on Xen Virtualization Platform”. Huazhong University of Science and Techonology, 2009.

Academic Service

Area Chair/Organizer

- Area Chair on ACL and ARR 2025

Program Committee Member/Reviewer

- Annual meeting of the Association of Computational Linguistics (ACL), 2020-25
- ACL Rolling Review, 2021-25
- Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020-25
- North American Chapter of the ACL (NAACL), 2021,22,24,25
- Conference of the European Chapter of the ACL (EACL), 2021
- International Conference on Computational Linguistics (COLING), 2020-25
- Conference on Computational Natural Language Learning (CoNLL), 2021-24
- International Conference on Artificial Intelligence in Education (AIED), 2023,2024
- International Conference on Educational Data Mining (EDM), 2024-25
- Conference on Language Modeling(COLM), 2024-25
- Conference of the Association for the Advancement of Artificial Intelligence (AAAI), 2020–25
- Shared Task on Cross-Framework Meaning Representation Parsing at the 2019 Conference on Natural Language Learning (MRP’2019@CoNLL)
- Workshop on Innovative Use of NLP for Building Educational Applications (BEA), 2023-25
- Workshop on NLP for Conversational AI (NLP4ConvAI), 2023
- Workshop on NLP for Indigenous Languages of the Americas (AmericasNLP), 2023
- Workshop on Speech and Language Technology in Education (SLaTE), 2023